Toward Faithful Scene-Adaptive Knowledge for Semantic Segmentation of Remote Sensing Images

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***Abstract*—Semantic segmentation is a critical procedure in remote sensing image analysis that supports various applications. High-resolution remote sensing images contain a wealth of ground object features, which are organized into various describable scenes. The visual content offset toward each scene is an intuitive sense for understanding geospatial objects. However, existing semantic segmentation methods for remote sensing images generally neglect this intuition and lack the ability to adjust their perception preference on different images. To address this problem, we propose a paradigm for collecting scene information and dynamically adjusting the model inference process to be scene-aware. Specifically, our method leverages the class feature from the image to enhance the fixed class representation from the model. The interaction of this information is facilitated by a neighbor-friendly embedding space, making it more faithful to associate the image features and model parameters. For the model to better understand the complex scenes, a manifold mixup method is proposed to expand the effective embedding space on intraclass and interclass regions, forcing the model to challenge the ambiguous instances in remote sensing images. Extensive experiments on four publicly available datasets demonstrated that our proposed method improved the accuracy of semantic segmentation models on remote sensing images, overcoming the state-of-the-art methods.**

***Index Terms*—Dynamic representation, manifold mixup, neighbor-friendly embedding space, semantic segmentation.**

I. Introduction

AUTOMATIC analysis of high-resolution remote sensing imagery plays a vital role in many applications that rely on spatiotemporal data, such as urban planning [1], [2], agriculture [3], disaster assessment [4], [5], and ecological monitoring [6], [7]. These images offer detailed spatial information about both natural landscapes and human-made structures on Earth's surface. The scenes [8], [9] captured in remote sensing data reflect a wide range of ground activities, represented by geospatial objects with similar features and clustered distribution. To extract meaningful statistics on these geospatial objects, there is a natural tendency for a remote sensing Earth observation to cover various scenes and heterogeneous geospatial objects, so that prejudices on monitoring the concerned geospatial phenomena can be alleviated [10]. Consequently, there is a need for semantic segmentation approaches in Earth observation to learn about complex scenes and adapt to varied environmental conditions present across different regions over the remote sensing images [11], [12].

Deep learning methods stand out at discovering intricate structures in high-dimensional data [13]. Over the past decades, they have driven significant progress in extracting discriminative features for the semantic segmentation of remote sensing images. Fully convolutional networks (FCNs) [14] have demonstrated strong capabilities in capturing texture and semantic features of each region in an image. However, the fixed receptive fields in convolution operations limit the model's ability to capture long-range dependencies, an essential factor for integrating contextual visual information [15], [16], [17]. Access to broader contextual cues allows the model to interpret a geospatial entity more accurately by considering its surrounding environment [18]. Despite these advances, the complex background information in each scene makes the existing methods difficult to alleviate the high intraclass and interclass ambiguities in recognizing diverse geospatial entities. To promote the distinguishing ability of different classes, some recent approaches have incorporated contrastive loss functions alongside traditional segmentation objectives to directly optimize the pixel-wise features or embeddings of the remote sensing images [19]. These loss functions encourage the model to draw features of the same class closer in the embedding space while pushing apart features of different classes, thereby enhancing its discriminatory power [20], [21], [22]. While these methods have led to noticeable improvements, they overlook two specific challenges in high-resolution remote sensing imagery as follows.

1) *Scene Variation:* The appearance of geospatial objects manifests significant bias toward different scenes. As shown in Fig. 1(a), each remote sensing image typically includes a set of similar objects whose visual features are influenced by the broader scene context. For example, the co-occurrence probability of yachts and freighters

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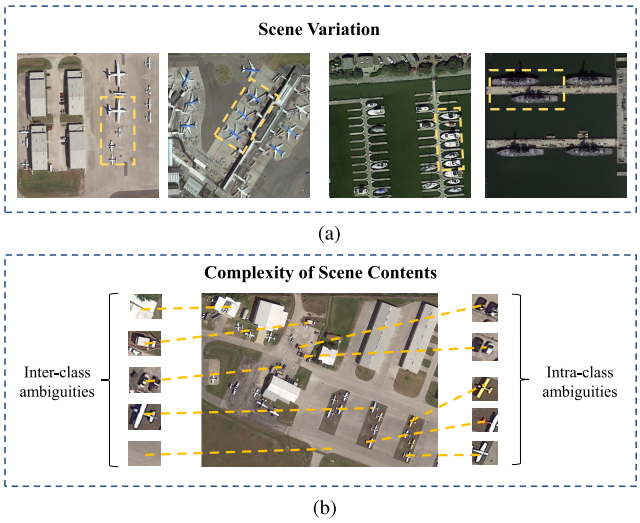

Fig. 1. Challenges in high-resolution remote sensing images: (a) scene variation, and (b) complexity of scene contents.

is usually lower than when they gather in separate sites, respectively. Previous methods typically rely on static classifiers that process all images uniformly, without adjusting the model inference behaviors explicitly according to the current scene information. This hinders the semantic segmentation methods from focusing on the unique scene information to perceive the geospatial objects flexibly.

2) *Complexity of Scene Contents:* The scene contents are perceived as numerous object parts in segmentation tasks by the visual systems, as shown in Fig. 1(b), which bears more interclass and intraclass ambiguities due to the diverse granularity of object parts [23]. Previous methods do not explore the ambiguous mixed representation between these geospatial objects in the embedding space, which is more difficult to classify. This hinders them from sufficiently capturing the information for each scene.

In view of the above-mentioned problems, we propose an embedding space organization and optimization model, named dynamic scene-adaptive network (DSA-Net), that can be integrated with various feature extractors to improve the performance of semantic segmentation models. DSA-Net seeks to enhance semantic segmentation models according to the characteristics of scene variation and complexity of scene contents in the remote sensing images. On the one hand, flexible class representations are contained in various scenes. It is beneficial to adjust the model inference behavior accordingly for the semantic segmentation of each image. On the other hand, the ambiguities among scene contents motivate us to generate challenging intraclass and interclass samples within the feature space and inject this knowledge into the model for better understanding each scene.

To achieve these goals, we first construct a neighbor-friendly embedding space[1] [24] that produces the classification probability based on Euclidean distance to facilitate the interaction between features of different geospatial objects. Second, to account for scene variation across remote sensing images, we propose to incorporate both the static prototypes[2] from the classifier and the dynamic prototypes from the image features. The neighbor-friendly embedding space unifies the representations of the static and dynamic prototypes to the same scale, guaranteeing that the combination of them is effective in describing the samples in the embedding space. Third, we propose to further inject the knowledge of classification ambiguities of the heterogeneous geospatial objects into the model through manifold mixup in the embedding space. This approach expands the effective embedding space of pixel representation, while the relations between mixed sample features and the class prototypes also retain their effectiveness in the neighbor-friendly embedding space.

To summarize, the main contributions of this article are as follows.

1) We propose a novel embedding space organization method that endows the semantic segmentation models with scene-adaptive recognition ability through a faithful incorporation of dynamic and static prototypes.
2) We propose to optimize the embedding space with augmented samples generated by manifold mixup methods, which enhance the models to better capture the intraclass and interclass representations across diverse geospatial instances for better understanding each scene.
3) Through extensive experiments, we evaluated the effectiveness of the proposed method on enhancing the accuracy of different semantic segmentation models, and demonstrated that our method outperforms the state-of-the-art approaches.

II. Related Work

A. Semantic Segmentation of Remote Sensing Imagery

FCNs [14] represent a foundational approach for end-to-end pixel-level classification in image segmentation. Subsequent advancements in image segmentation methods mainly focused on refining multiscale feature fusion [25], [26], [27], [28], [29], and preserving high-resolution spatial details [30] for capturing the fine structures of objects. Global information about class representation is also introduced for refining the extracted features [31]. More recent approaches model relational semantic features for comprehensively parsing an image [32], [33], [34].

Various attempts have been made to improve the performance of semantic segmentation on remote sensing images in the past few years, which can be roughly divided into two main directions. The first direction is to focus on more sophisticated feature engineering of the spatial texture. Representative methods on this direction include multiscale or coarse-to-fine feature fusion [35], [36], [37], [38], [39], [40], regional attention on features [41], boundary-sensitive features [42], [43], and resolution-preserving features [44]. Although achieving

[1]In this article, the term "embedding space" refers to the space of input features to the final classifier of the semantic segmentation models.

[2]A prototype is the center of all samples of the same class in the embedding space. As a representer of the class information, the relations between samples and prototypes are closely related to the classification result.

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encouraging performance, these networks have limited receptive fields and are unable to capture global information from the whole image. To solve this problem, the second direction focuses on incorporating long-range information from the RS images. Techniques in this category include dilated or large-kernel methods [45], spatial long-range features [17], [46], [47], [48], [49], and collaborative local–global architectures [50], [51], [52]. Transformer-based architectures, which rely on self-attention mechanisms, have also been adopted for remote sensing tasks due to their capacity to model long-distance relationships across feature tokens. Recent models on remote sensing image segmentation combine Transformer features with CNN-based backbones to harness both local and global cues [16], [53], [54], [55], [56], [57], while others focus on learning meaningful class token representations to improve segmentation accuracy [58]. Some latest works consider more detailed features such as frequency domain information [59], special filters for boundary refinement [60], frequency-domain sparse feature expression [61], or special output correcting tokens [62] to improve the feature description capability of Transformer architectures for remote sensing image semantic segmentation.

B. Prototype-Based Metric Learning

In metric learning, a prototype is also known as a center [63] or a proxy [64] that represents the central feature vector for a group of related examples. The relation between a prototype and an example is typically measured using metrics such as the Euclidean distance, cosine similarity, or other learnable metrics. A prototype can be learned or extracted from the samples, as commonly done in the few-shot classification or segmentation models. The estimation of a prototype takes into account the relationships between the prototype and a subset of examples, which captures the structure of parts of an embedding space. In the context of remote sensing segmentation, RSProtoSeg [22] proposes to learn static prototypes with fine-grained within-class patterns. MPCNet [58] proposes a dynamic class token update method for combining the pixel features in each remote sensing image. However, the class token update through attention operations is complicated and does not encode the global scene information, thus making the classifier suboptimal. Our method represents the static prototypes and dynamic prototypes for each scene in an explicit and easy-to-interact-with relationship, thus better discerning the scene-adaptive knowledge for classification and potential analysis.

C. Mixup

The utility of the mixup techniques has been demonstrated by numerous articles on robust representation learning, alleviating failure under distribution shift, and improving in-distribution accuracy [65], [66], [67]. Mixup is suggested to assist representation learning by supplementing the binary-value classification label (1 for positive and 0 for negative) with a data interpolation ratio λ between 0 and 1 in methods such as *i*-Mix [68] and Un-Mix [69]. Besides input mixup [70] that interpolates examples in the input space, there are manifold mixup [71] that linearly interpolates intermediate representations instead, so that the structure of the representation space is learned. The mechanism of the m^2-Mix manifold mixup method is to expand effective embedding spaces that connect heterogeneous domains, increasing the uniformity of multimodal representations [72]. Our method focuses on the diversity of the geospatial objects and the interaction between the representations of different classes. The manifold mixup approach helps the model learn continuous transitions between geospatial object categories, thereby improving its ability to resolve intraclass and interclass ambiguities in complex remote sensing scenes.

III. Method

A. Overview

As shown in Fig. 2, given an input image I, the encoder first transforms it into a feature map $X \in \mathbb{R}^{H\times W\times C}$ with its height H, width W, and number of channels C. The feature map is embedded into a metric space learned based on the nearest neighbor rules with the help of the prototype layer. A prototype layer $\theta \in \mathbb{R}^{K\times C}$, where K is the number of classes in the semantic segmentation task, stores the static prototypes for yielding the coarse classification mask $\tilde{M} \in \mathbb{R}^{H\times W\times K}$. The dynamic prototypes extracted from the input images are combined with the static prototypes to form the scene-adaptive prototypes, which then yield the final classification masks $M \in \mathbb{R}^{H\times W\times K}$. The scene-adaptive prototypes extend the static prototypes to track the context on every input image, making the classifier more flexible in handling different images. To further enhance the model's capability of capturing the intraclass and interclass relationships across diverse geospatial samples, a manifold mixup approach is proposed for generating augmented samples in the embedding space. Overall, the loss functions for training DSA-Net include cross-entropy (CE) objectives on the coarse and final semantic segmentation results, $L_{\text{coarse-CE}}$ and $L_{\text{final-CE}}$, and on the intraclass and interclass mixed samples, $L_{\text{intraclass}}$ and $L_{\text{interclass}}$. The overall loss function is as follows:

$$L = L_{\text{coarse-CE}} + L_{\text{final-CE}} + \mu_1 L_{\text{interclass}} + \mu_2 L_{\text{interclass}}. \quad (1)$$

B. Learning a Steerable Embedding Space

A classifier $f : \mathcal{X} \to \mathcal{Y}$ maps a linear space of features $\mathcal{X} \subset \mathbb{R}^C$ into a linear space of class-specific scores $\mathcal{Y} \subset \mathbb{R}^K$. We choose to embed each pixel of the input images into a metric space equipped with Euclidean distance learned based on the nearest neighbor rule, which is achieved by employing the prototype layer as the classifier on top of the encoder. A distance-based classifier is different from a linear classifier in the process of yielding the class-specific scores. A linear classifier gives the class-specific score for each class k as

$$f_k(x; w) = w_k^T x \quad (2)$$

where x is a C-dimensional input vector, and $w_k \in \mathbb{R}^C$ is the weight parameter on class k. Previous works on prototype networks indicate that a Euclidean distance-based classifier can be recognized as a linear classifier with parameterized

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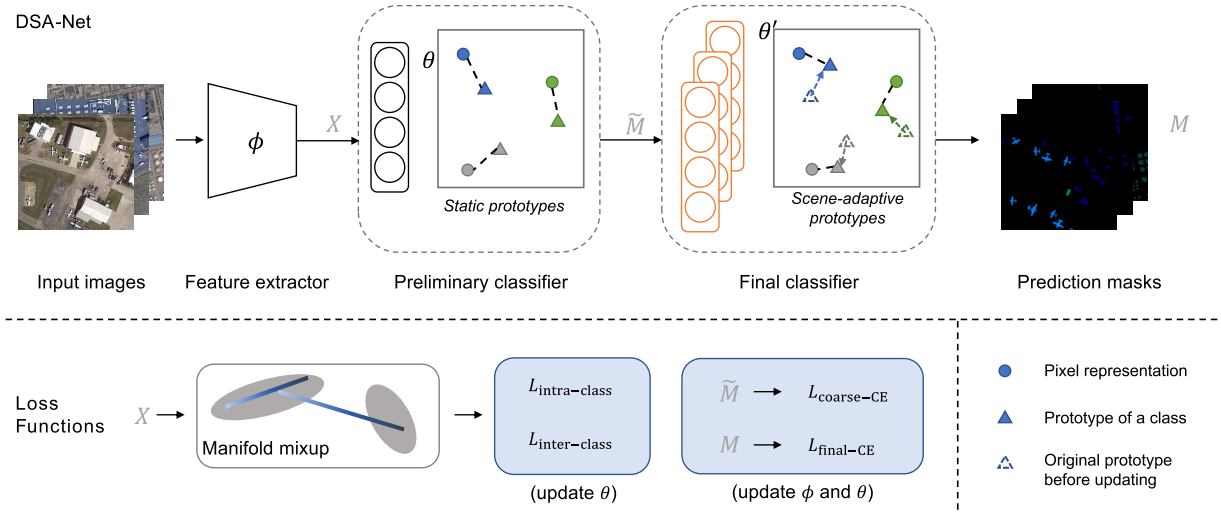

Fig. 2. Overview of the proposed DSA-Net and the training approach. The embedding space is organized by a learnable prototype layer based on the nearest-neighbor rule measured by the Euclidean distance for classification. Scene-adaptive dynamic prototypes are extracted from the image features for updating the class prototypes. Manifold mixup generates augmented samples that expand the effective embedding space for the network to better perceive the ambiguities between different geospatial entities.

bias [73], [74]. However, as we found in experiments, training a linear classifier will lead to a large disparity between the magnitudes of the classifier weights w_k and the input feature x, resulting in difficulty in combining the static classifier parameters and the dynamic image-specific prototypes. In Table I, a series of training results on the linear classifiers and their variants with constraints on the norms of their weights and input features are displayed. We make the following observations from these results.

1) If the linear classifier is used, the norms of the classifier weights and the input features will have a large difference on the final training states, as shown in cases 1–3.
2) Forcing both the norms of weights and input features to be identical leads to a significant performance drop, as shown in case 4.

From points 1) and 2), it can be known that the magnitudes of the classifier parameters and the input features in the metric space contain necessary information for accurate classification. Therefore, it is beneficial to: 1) retain the magnitudes of the classifier parameters and the input features, so that the effective class information is preserved in them; and 2) make the classifier parameters and input features distribute on a unified scale, so that they can be effectively integrated. To achieve these goals, we propose to use the prototype layer to learn a metric space for Euclidean distance classification.

The prototype layer computes the class-specific score for class k as follows:

$$f_k(x;\theta) = -\frac{1}{C}\|\theta_k - x\|^2 \tag{3}$$

where $\theta_k \in \mathbb{R}^C$ is the kth learnable prototypes in the prototype layer, and $\|\cdot\|$ is l_2-norm. The prototype layer determines the class of the input instance based on its distances to each classification prototype. The learning of the prototypes through (3) leads to an estimation of the centers of the samples of each class in the embedding space. Therefore, the learned static prototypes turn out to have their norms close to the image pixel features.

Remark 1: Denote $x \sim \mathcal{D}$ as the process of drawing samples from the data distribution $\mathcal{D}$ in space $\mathcal{X}$. Denote η_x is a coefficient that influences the update step-size of the parameters of the classifier, which is related to the sample x according to the derivation of CE loss functions on softmax prediction results. Then the classifier parameter update steps from iteration t to $t+1$ produced by (2) and (3) are listed, respectively, as follows:

$$w^{t+1} = w^t - \mathbb{E}_{x\sim\mathcal{D}}\left[\eta x^t\right] \tag{4}$$

$$\theta^{t+1} = \mathbb{E}_{x\sim\mathcal{D}}\left[(1-\eta)\,\theta^t + \eta x^t\right] \tag{5}$$

and we denote η_{x^t} as η for simplicity.

Proof: Suppose the class-specific scores $f(x;\theta)$ parameterized by θ is supervised by a loss function $L(f)$ for optimization. Based on the chain rule of derivatives, the gradient for updating θ is

$$\nabla_\theta L = \frac{\partial L}{\partial f}\nabla_\theta f.$$

The gradient of f with respect to its parameters is inferred from (2) and (3) are,

$$\nabla_w f(x;w) = x, \text{ and } \nabla_\theta f(x;\theta) = \frac{2}{C}(x-\theta)$$

respectively. Combining the gradient update step $\theta^{t+1} = \theta^t - \nabla_\theta L$ with the above representations yields (1) and (5).■

It can be seen from the above update steps that the values of the feature x plays different roles in the gradient descent optimization of the linear classifier and the prototype classifier. As for the former, x becomes the shifts of classifier parameters; while for the latter, x is a target object of classifier parameters

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Algorithm 1 Semantic Segmentation With Scene-Adaptive Prototypes

Input: Input image I, static prototypes θ
Output: Semantic segmentation mask M

1: Get image feature X
2: Get coarse mask $\tilde{M}$ through (3) with θ
3: Get dynamic class representation p from image feature through (6)
4: Update to scene-adaptive prototypes θ' through (7)
5: Get final mask M through (3) with θ'

in the momentum update. This indicates that the optimization processes of the linear classifiers and the prototype-based classifiers are different and need specific considerations on how to organize the embedding spaces.

C. Scene-Adaptive Prototypes

As shown in Fig. 2, the static prototypes θ are fixed for all images, and our goal is to update the static prototypes into scene-adaptive prototypes $\theta'^{(i)}$ conditioned on each image data $X^{(i)}$, with i referring to the ith image in a batch. To achieve this, a guiding semantic segmentation mask $G^{(i)} \in \mathbb{R}^{H\times W\times K}$ for this image is adopted. It provides classification probabilities on each pixel that can then be used to calculate the dynamic prototypes from the image features, which is formulated as

$$p_k^{(i)} = \text{Mean}\left(X^{(i)} \odot \text{softmax}_{HW}\left(G_k^{(i)}\right)\right) \tag{6}$$

where $\odot$ is the element-wise multiplication, the softmax_{HW} operation is applied on the dimension of height and width ($H \times W$) of the image features, and Mean$(\cdot)$ computes the average weighted by $\text{softmax}_{HW}(G_k^{(i)})$. In the training phase, G is chosen as the ground truth semantic segmentation mask after one-hot encoding. In the test phase, G is chosen as the predicted pixel classification probability, i.e., $G_k^{(i)} := \text{softmax}_K(\tilde{M}_k^{(i)})$, where the softmax_K operation is applied on the dimension of classes.

Finally, the scene-adaptive prototypes θ' are generated from the static prototypes and the dynamic prototypes with a learnable projection layer ψ that combines the semantic information across the channels of both kinds of prototypes, which is formulated as

$$\theta_k'^{(i)} = \begin{cases} \psi\left(\left[p_k^{(i)}, \theta_k^{(i)}\right]\right), & \text{if } k \in \mathcal{C}_{\text{top-1}}\left[\tilde{M}^{(i)}\right] \\ \theta_k^{(i)}, & \text{if } k \notin \mathcal{C}_{\text{top-1}}\left[\tilde{M}^{(i)}\right]. \end{cases} \tag{7}$$

Here, $\mathcal{C}_{\text{top-1}}[\tilde{M}] = \{k | k \in \arg\max_K \tilde{M}\}$ is the operation of getting all unique elements included in the class indices with top-1 class-wise probability $\tilde{M}$ for each pixel. This hard constraint ensures that the prototype update only happens on classes appearing in the guiding semantic segmentation mask G. It excludes low-confident pixel classification judgments that could potentially incur unreliable information to the dynamic prototypes $p_k^{(i)}$. The process of semantic segmentation with scene-adaptive prototypes is summarized in Algorithm 1.

To train the scene-adaptive prototype update module, we impose supervision on both the coarse and the final semantic segmentation prediction results. Denote L_{CE} the CE function and Y the ground truth semantic segmentation mask. The corresponding loss functions are $L_{\text{coarse-CE}} := L_{\text{CE}}(\tilde{M}, Y|\phi, \theta)$, and $L_{\text{final-CE}} := L_{\text{CE}}(M, Y|\phi, \theta)$.

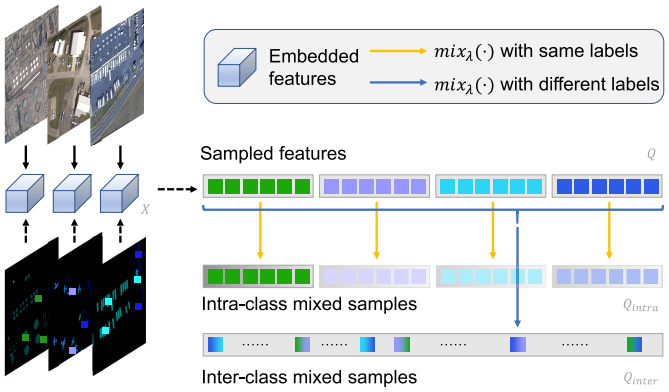

Fig. 3. Overview of the workflow of intraclass and interclass mixup.

D. Manifold Mixup for Capturing Class Relations

Manifold mixup generates new samples in the embedding space directly. Given a pair of pixel features, $x, x' \in \mathcal{X}$, the manifold mixup is formulated as

$$\text{mix}_\lambda\left(x, x'\right) = \lambda x + (1-\lambda)\, x' \tag{8}$$

where the interpolation factor λ is a random variable drawn from uniform distribution on $[0, 1]$. For each pixel pair to be mixed, a random interpolation factor is generated independently.

In the training phase of our model, provided two samples in the embedding space, $x_1, x_2 \in \mathcal{X}$, with their ground truth classification label $y_1, y_2 \in \mathcal{Y}$, we generate a mixed sample $\hat{x}$ with its mixed label $\hat{y}$ as $\hat{x} = \text{mix}_\lambda(x_1, x_2)$, and $\hat{y} = \text{mix}_\lambda(y_1, y_2)$. The specific workflow of the manifold mixup process in our proposed method is shown in Fig. 3. We sample pixel embeddings from each image in a batch and assemble them together into a query Q of sampled features shared by the current batch, formulated as

$$Q = \{(x, y)\,|x \sim U\left(X\left[\mathbb{1}_{Y=k}\right]\right), y = k, k \in [0, \ldots, K-1]\} \tag{9}$$

where $\mathbb{1}_{Y=k}$ is the indexing mask for all pixels of class k, and $U(\cdot)$ is the uniform sampling operation that draws a fixed number of pixels. Finally, intraclass mixup and interclass mixup are carried out on the sampled feature query, yielding queries Q_{intra}, Q_{inter} with the same length as Q, containing the intraclass and interclass mixed feature and label pairs $(\hat{x}, \hat{y})$, respectively. The process of generating manifold mixup samples is summarized in Algorithm 2.

The generated samples Q_{intra} and Q_{inter} are applied to train the prototype layer θ with CE loss functions on the generated mixed samples. Through experiments, we found that directly training with the class-wise uniformly drawn samples did not produce good results. Therefore, we reweigh the CE loss functions on each class with the pixel occurrence distribution on the current training batch. Denote the number of pixels for the class k in the current batch as $N_k = \sum \mathbb{1}_{Y=k}$ such that

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Algorithm 2 Generate Manifold Mixup Samples

Input: Image feature X, Ground truth semantic segmentation mask Y

Output: Intra-class mixed samples Q_{intra}, Inter-class mixed samples Q_{inter}

1: Randomly select an equal number of pixels for each class with Y
2: Get the features of these pixels from X, and arrange the pixel features and labels into a queue Q through (9)
3: Perform intra-class mixup on Q for each class, and get the queue of intra-class mixed samples Q_{intra}
4: Perform inter-class mixup on Q, and get the queue of inter-class mixed samples Q_{inter}

the weight for class k is $w_k = N_k / \sum_{k=0}^{K-1} N_k$, the weighted CE function is

$$L_{\text{weighted-CE}}(X, Y, w) = -\sum_{x \in X} \sum_{y \in Y} \sum_{k=0}^{K-1} w_k y_k \log \sigma(f_k(x)) \tag{10}$$

where $f_k(x)$ is the predicted score of class k, and σ is the softmax function. Thus, the loss functions for learning the intraclass and interclass mixed samples are $L_{\text{intraclass}} := L_{\text{weighted-CE}}(Q_{\text{intra}}, w|\theta)$ and $L_{\text{interclass}} := L_{\text{weighted-CE}}(Q_{\text{inter}}, w|\theta)$.

IV. Experiments

A. Datasets

1) iSAID: The iSAID dataset [79] stands as one of the largest geospatial semantic segmentation datasets for remote sensing imagery, containing 655 451 densely annotated object instances across 15 categories within 2806 high-resolution images. The images of iSAID are selected under multiple resolutions and captured from multiple sensors and platforms. The 15 categories of geospatial objects contained by the iSAID dataset are ship (SH), storage tank (ST), baseball diamond (BD), tennis court (TC), basketball court (BC), ground field track (GTF), bridge (BR), large vehicle (LV), small vehicle (SV), helicopter (HC), swimming pool (SP), roundabout (RA), soccer ball field (SBF), plane (PL), and harbor (HB). The original image sizes range from 800×800 pixels to $4000 \times 13\,000$ pixels in the dataset, and all images are divided into training, validation, and test sets with 1411, 458, and 937 images, respectively. We use the validation set for evaluation because the ground truth labels of the test set are unavailable.

2) Vaihingen: The ISPRS Vaihingen dataset [80] consists of 33 very-high-resolution (VHR) true orthophoto aerial image with 9 cm/pixel spatial resolution collected over 1.38 km^2 area of the Vaihingen city. The average size of the images is 2494×2064 pixels. Each image is composed of three bands: near infrared (NIR), red (R), and green (G). The dataset involves six categories in its ground truth labels: impervious surface (Imp.), building (Build.), low vegetation (L. veg.), tree, car, and clutter. The clutter category is ignored in our experiments. Following the official splits, we utilize 17 images, IDs: 2, 4, 6, 8, 10, 12, 14, 16, 20, 22, 24, 27, 29, 31, 33, 35, and 38, for testing, and the remaining 16 images for training. For evaluation, we use the ground truth labels with eroded boundaries provided by the official data, which ignores a circular disk of 3−pixel radius around the boundaries of each semantic region when evaluating the segmentation results.

TABLE I
Influence of the Norms of Classifier Weights and Input Features Onlinear Classifiers

Case		Accuracy (mIoU)	Average norm of classifier parameters	Average norm of input features
1	Linear classifier	0.6801	0.98	18.19
2	+ norm-1 weights	0.6824	1	20.01
3	+ norm-1 input features	0.6838	8.31	1
4	+ both	0.6268	1	1

3) Potsdam: The ISPRS Potsdam dataset [80] includes 38 VHR true orthophoto aerial images with a spatial resolution of 5 cm/pixel that cover an area of 3.42 km^2 of the Potsdam city. The size of each image is 6000×6000 pixels. Four bands are provided in the multispectral image data of the dataset: near infrared (NIR), red (R), green (G), and blue (B). We use the RGB bands for experiments. The semantic categories of the Potsdam dataset are the same as the Vaihingen dataset. Following the official splits, we utilize 24 images for training, and the remaining 14 images for testing, IDs: 2_13, 2_14, 3_13, 3_14, 4_13, 4_14, 4_15, 5_13, 5_14, 5_15, 6_13, 6_14, 6_15, and 7_13. For evaluation, similar to the Vaihingen dataset, we also use the ground truth labels with 3 − pixel eroded boundaries in the official data to reduce the uncertainty caused by the border segmentation error.

4) LoveDA: The LoveDA dataset [81] is a challenging remote sensing semantic segmentation dataset including images in both rural and urban areas of three cities, covering a total area of over 536 km^2 with a spatial resolution of 0.3 m/pixel. Seven different land cover categories are included in the dataset: buildings (Build.), roads, water, barrenland, forests, agriculture (Agri.), and background (BG). The dataset are split into 2522 images for training, 1669 images for validation, and 1796 for testing. The evaluation results are acquired on the test set through the official online test platform.

B. Evaluation Metrics

To assess the quantitative performance, three commonly used metrics for semantic segmentation tasks, overall accuracy (OA), mean $F1$-score (m$F1$), and mean intersection over union (mIoU), are employed in our experiments.

C. Implementation Details

Our model is implemented by the PyTorch library and trained on Tesla V100 GPUs. For the iSAID dataset, we crop the image into 896×896 with stride 512 following [12], [17] for training. For the Vaihingen and the Potsdam datasets, the input image is cropped to 512×512 with 50% overlap [58]. For the LoveDA dataset, the patch size is set to 512×512 for training. The ADAM optimizer is used to update model parameters with an initial learning rate of 0.0001 and a weight decay of 0.01. A polynomial learning rate policy is adopted

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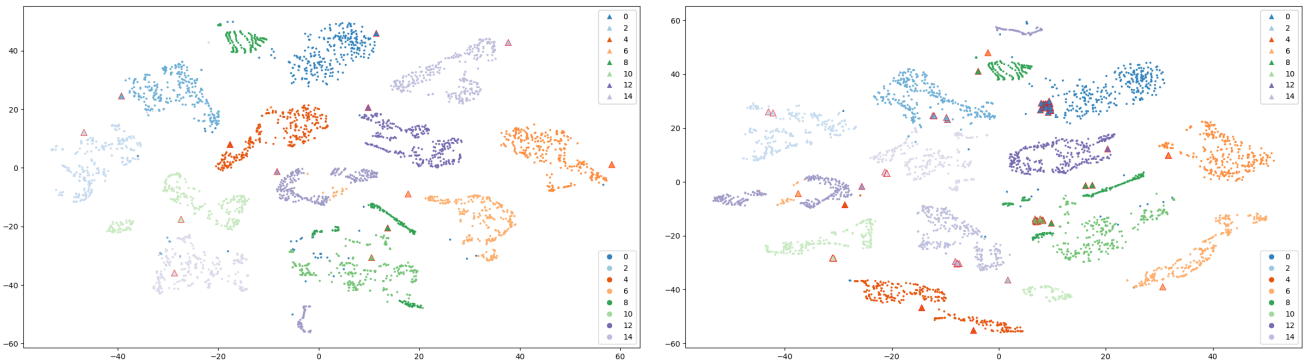

(a) (b)

Fig. 4. Visualization of the pixels and prototypes in the embedding space $\mathcal{X}$ through t-SNE for dimensionality reduction. Circles indicate pixel embeddings. Triangles indicate prototype embeddings, which are specifically divided into (a) static prototypes and (b) scene-adaptive dynamic prototypes.

TABLE II
QUANTITATIVE EVALUATION RESULTS ON THE VALIDATION SET OF iSAID DATASET

Method	Backbone	IoU per category (%)															mIoU (%)
		SH	ST	BD	TC	BC	GTF	BR	LV	SV	HC	SP	RA	SBF	PL	HA	
DenseASPP [75]	ResNet-50	55.70	63.50	67.20	81.70	54.80	52.60	34.70	55.60	36.30	33.40	37.50	53.40	73.30	74.70	46.70	57.30
RefineNet [76]	ResNet-50	63.80	58.60	72.30	85.30	61.10	52.80	32.60	58.20	42.40	23.00	43.40	65.60	74.40	79.90	51.10	60.20
FarSeg [12]	ResNet-50	67.17	69.66	77.13	89.88	63.87	47.02	37.65	63.26	50.48	35.02	50.40	65.87	73.36	85.03	57.92	64.55
PFSegNet [17]	ResNet-50	72.70	74.90	71.20	88.30	63.00	**62.30**	35.00	64.80	51.80	41.30	52.50	63.30	74.40	85.10	59.50	66.20
SCO [21]	ResNet-50	**73.37**	75.75	79.61	89.93	68.17	61.19	39.38	65.09	52.34	36.95	51.67	67.97	**76.08**	85.82	**62.20**	67.79
MPCNet [58]	ResNet-50	69.90	70.70	78.60	89.60	**70.00**	58.70	40.20	62.60	51.70	**41.70**	51.30	64.20	74.30	86.20	56.70	66.60
SegCLIP [11]	ResNet-50	70.45	68.06	66.90	86.93	58.55	59.23	**44.82**	65.22	53.50	32.17	46.76	**75.58**	74.29	85.43	59.17	65.39
LOGCAN++ [77]	ResNet-50	70.81	75.30	**81.48**	90.08	69.20	47.85	40.63	64.43	52.72	39.60	47.46	62.96	72.14	**86.50**	59.63	66.24
UNetFormer [16]	ResNet-50	70.00	71.24	78.03	89.67	69.79	57.87	40.30	61.79	50.82	34.83	**54.47**	64.72	75.21	85.56	57.43	66.30
DSA-Net	ResNet-50	72.47	**76.61**	79.54	**90.30**	67.51	59.00	39.60	**66.02**	**55.51**	40.03	53.78	71.94	75.47	86.41	59.16	**68.28**
FarSeg [12]	ResNet-101	69.34	72.14	81.90	90.37	65.16	58.37	38.98	63.56	50.25	38.80	49.12	55.87	74.70	85.86	56.71	65.64
PFSegNet [17]	ResNet-101	69.77	73.33	82.03	89.12	**70.38**	60.20	37.58	61.89	55.46	27.07	53.13	65.00	74.31	85.80	59.83	66.50
SCO [21]	ResNet-101	**74.16**	75.51	80.06	90.69	68.68	60.28	38.14	64.79	53.49	42.51	53.01	71.80	76.11	85.97	60.44	68.42
MPCNet [58]	ResNet-101	72.77	74.44	82.89	89.86	67.84	60.53	**43.00**	65.76	53.18	40.48	53.78	74.13	75.79	86.47	61.19	68.83
CMTFNet [78]	ResNet-101	65.02	71.67	77.01	87.62	61.37	62.71	40.19	59.98	43.68	16.94	47.35	72.74	**76.71**	80.48	55.49	63.63
LOGCAN++ [77]	ResNet-101	68.79	75.36	79.94	88.41	64.16	55.05	41.81	64.20	53.48	42.09	53.68	62.58	75.05	86.93	60.96	66.98
UNetFormer [16]	ResNet-101	70.74	74.58	79.50	90.52	69.23	58.17	36.61	63.93	52.80	41.73	54.03	**74.61**	75.50	86.72	58.75	67.91
DSA-Net	ResNet-101	72.82	**76.29**	**83.51**	**90.78**	65.99	**65.96**	41.67	**66.76**	**55.94**	**46.01**	**56.11**	71.91	76.63	**87.22**	**62.77**	**69.97**
OCR [33]	HRNet-w48	67.60	75.00	78.50	88.50	57.40	59.60	**45.30**	63.40	49.20	35.70	46.40	66.70	72.30	84.10	55.00	63.00
RSProtoSeg [22]	HRNet-w48	72.28	73.57	80.17	88.68	62.72	60.29	37.08	64.70	51.69	38.24	45.75	65.95	70.38	84.91	60.01	65.90
DSA-Net	HRNet-w48	**75.42**	**75.50**	**84.40**	**89.77**	**66.46**	**64.45**	40.64	**66.81**	**57.47**	**40.90**	**51.33**	**75.68**	**76.59**	**87.37**	**65.94**	**69.87**
MPCNet [58]	Swin-B	76.50	75.60	83.40	89.40	**71.70**	62.50	41.10	65.80	53.90	**48.10**	55.80	72.20	76.50	86.70	62.40	70.20
SegCLIP [11]	Swin-B	73.48	74.24	68.98	87.66	62.49	**65.49**	**45.55**	**68.20**	54.27	38.61	44.99	**74.88**	**79.11**	86.60	62.83	67.78
MeSAM [62]	Swin-B	73.55	76.40	82.81	89.88	67.70	61.25	42.30	66.40	53.50	43.83	54.92	61.72	74.42	86.08	60.27	68.39
GF-ResFormer [60]	Swin-B	73.45	73.54	80.57	90.34	65.62	61.12	39.01	64.74	54.96	45.81	52.69	68.09	74.66	**87.09**	59.95	68.17
DSA-Net	Swin-B	**76.66**	**77.82**	**84.23**	**90.90**	69.76	62.77	39.42	67.13	**55.56**	45.44	**55.95**	73.50	76.36	87.06	**63.38**	**70.32**

as $\mathrm{lr} = \mathrm{lr}_0 \times (1 - i/T)^{0.9}$, where lr_0 is the initial learning rate, i denotes the current number of iterations, and T denotes the total number of iterations. We empirically set $\mu_1 = \mu_2 = 0.3$ for training, and we train the model without the manifold mixup loss functions in the last quarters of training iterations to make the training more stable. To avoid overfitting during training, we adopt the following data augmentation transforms: random resize (with a scale factor from 0.5 to 1.5), random crop, random flip, and random 90° rotations. The model is trained for 80-k iterations on the iSAID dataset, 8-k iterations on the Vaihingen dataset, 30-k iterations on the Potsdam dataset, and 50-k iterations on the LoveDA dataset. The batch size is set to 8.

D. Comparison to General Methods and Result Demonstration

1) Experiments on the iSAID Dataset: To verify the effectiveness of our proposed DSA-Net, we first conducted the experiments on the large-scale iSAID dataset for a comparison with the state-of-the-art methods. For the purpose of a fair comparison, we applied the proposed classifier and training approach on different backbones as the encoder. Table II reports the per-class IoU and the overall mIoU on the iSAID benchmark. It can be observed that our method surpasses other methods under different backbone settings. The ResNet-50 and ResNet-101 [82] backbones are commonly used feature extraction convolutional neural networks, and DSA-Net shows

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TABLE III
QUANTITATIVE EVALUATION RESULTS ON THE TEST SET OF ISPRS VAIHINGEN DATASET

Method	F1-score					mF1	OA	mIoU
	Imp.	Build.	L.veg.	Tree	Car			
FCN [14]	93.26	96.07	85.43	90.64	86.91	90.46	91.49	82.82
FPN [29]	93.20	95.90	85.28	90.62	87.49	90.50	91.39	82.86
PSPNet [27]	92.68	95.79	83.46	89.47	87.15	89.71	90.61	81.62
Deeplabv3+ [28]	93.13	96.13	85.10	90.39	88.36	90.62	91.38	83.07
FarSeg [12]	93.36	96.28	85.20	90.36	87.42	90.52	91.48	82.93
HMANet [49]	93.34	96.27	**85.59**	**90.67**	88.28	90.83	91.63	83.42
PFSegNet [17]	93.32	96.38	85.49	90.61	88.05	90.77	91.61	83.33
SCO [21]	93.08	96.28	84.16	90.00	88.48	90.40	91.16	82.74
UNetFormer [16]	93.37	96.12	84.99	90.44	88.49	90.68	91.44	83.18
MPCNet [58]	93.31	96.44	85.23	90.54	88.91	90.89	91.58	83.52
LOGCAN++ [77]	93.23	96.50	84.67	90.40	89.14	90.79	91.45	83.37
DSA-Net	**93.44**	**96.51**	85.38	90.60	**89.58**	**91.10**	**91.69**	**83.87**

TABLE IV
QUANTITATIVE EVALUATION RESULTS ON THE TEST SET OF ISPRS POTSDAM DATASET

Method	F1-score					mF1	OA	mIoU
	Imp.	Build.	L.veg.	Tree	Car			
FCN [14]	91.69	96.45	87.63	89.15	94.71	91.93	90.40	85.23
FPN [29]	92.97	96.99	88.37	89.30	96.28	92.78	91.26	86.74
PSPNet [27]	92.02	96.35	86.50	88.62	95.67	91.83	90.29	85.13
Deeplabv3+ [28]	93.21	96.93	88.13	89.48	96.25	92.80	91.33	86.77
FarSeg [12]	92.22	96.28	87.52	89.04	94.17	91.85	90.58	85.09
HMANet [49]	**94.44**	97.20	87.92	89.22	96.54	93.06	91.80	87.27
PFSegNet [17]	93.03	96.93	88.09	89.79	96.13	92.79	91.29	86.75
SCO [21]	93.48	97.15	87.74	89.46	96.56	92.88	91.38	86.93
UNetFormer [16]	92.94	96.94	87.91	89.55	95.89	92.64	91.20	86.49
MPCNet [58]	94.20	97.23	88.04	89.41	**96.62**	93.10	91.75	87.32
LOGCAN++ [77]	92.62	96.63	87.92	89.49	94.69	92.27	90.99	85.81
DSA-Net	94.15	**97.49**	**88.47**	**89.88**	96.52	**93.30**	**91.97**	**87.65**

TABLE V
QUANTITATIVE EVALUATION RESULTS ON THE TEST SET OF LOVEDA DATASET

Method	BG	Build.	Road	Water	Barren	Forest	Agri.	mIoU
FarSeg [12]	45.18	56.55	56.92	79.78	14.96	46.59	62.50	51.78
HMANet [49]	45.87	57.04	58.44	81.14	22.45	57.69	64.11	53.82
PFSegNet [17]	46.88	60.49	60.04	81.54	16.12	**50.22**	65.92	54.46
SCO [21]	46.95	59.46	**60.26**	79.81	16.02	48.84	66.89	54.03
UNetFormer [16]	46.53	61.00	57.00	81.88	**23.09**	48.28	65.69	54.78
MPCNet [58]	47.45	**61.57**	59.35	81.68	20.37	48.72	65.14	54.90
LOGCAN++ [77]	47.37	58.38	56.46	80.05	18.44	47.91	64.80	53.35
DSA-Net	**48.49**	60.03	54.87	**81.96**	22.98	48.91	**69.33**	**55.22**

TABLE VI
COMPARISON WITH OTHER CLASSIFIERS

Method	Params (M)	FLOPs (G)	FPS	Memory (GB)	mIoU
ConvNeXt	50.39	465.94	65.04	5.5	52.3
+ CenterSeg	50.39	467.37	23.53	5.6	52.5
+ MPC-Net	54.25	549.95	42.11	3.7	52.9
+ DSA-Net	50.43	466.34	40.61	5.6	53.4

competitive performance on the overall mIoU performance on these backbones. The high-resolution network HRNet-w48 [30] features the preservation of intermediate feature resolutions, and it manifests relatively higher accuracy on categories with complex shapes or boundaries, typically, such as the plane, harbor, and small vehicle objects. The Transformer-based architecture, Swin-B [83] backbone, reaches higher accuracies compared with other frequently used backbones. The proposed method is also applicable to the Swin-B backbone and promotes the performance to surpass the MPCNet, while MPCNet is harder to train on the more economic ResNet-50 backbone, and DSA-Net achieved a significantly superior result on this setting in comparison. These results show the ease of use of our proposed DSA-Net. Roughly speaking, it is a good idea to preserve a concise data flow for inference. From this perspective, it is a disadvantage of the SCO method to inflate its scales of input images by patch shuffling transformations without any evident influence on the pixel-averaged prototype representations. Arguably, treating the background class and the other foreground classes in significantly different operations, such as whether or not to use k-means clustering on the background or foreground classes in SCO, is also not a cost-effective choice for learning a model. As shown in Fig. 4, although there may be some degree of differences between the representations of background and foreground classes, the proposed DSA-Net can learn to update the prototypes according to the scene diversity adaptively, while designating the representation diversity for different classes a priori may be harmful for learning the model.

2) Experiments on Other Datasets: We conducted experiments on the test sets of the ISPRS Vaihingen dataset, the ISPRS Potsdam dataset, and the LoveDA dataset. For the Vaihingen and Potsdam datasets, the evaluation results of per-class and OA are presented in Tables III and IV. For a fair comparison, we employ ResNet-101 [82] as the backbone for all methods, and use the same setting on the inference process of the test images. The proposed DSA-Net achieved superior performance compared with other semantic segmentation methods. Specifically, compared with the multiscale segmentation networks such as feature pyramid network (FPN) [29] and FarSeg [12], DSA-Net exhibits a significant performance improvement with about 1% on the mIoU metric. On the networks that propagate or features, such as PFSegNet [17] and UNetFormer [16], DSA-Net achieves better performance on both the less heterogeneous Vaihingen dataset and the Potsdam dataset, verifying the effectiveness of leveraging the scene information to dynamically update the model. In addition, DSA-Net outperforms MPCNet [58] with a simpler structure of the classifier, indicating the potential benefits of applying the global scene feature in a specific embedding space. Compared with the Vaihingen and Potsdam datasets, the LoveDA dataset contains more diverse urban and rural scenes for parsing. The experimental results on the LoveDA dataset are shown in Table V with all networks employed, ResNet-101 as the backbone. DSA-Net achieves competitive performance among the commonly used remote sensing image semantic segmentation methods under more complex geography contexts.

3) Comparison With Other Classifiers: Table VI shows the performance and accuracy results of equipping special

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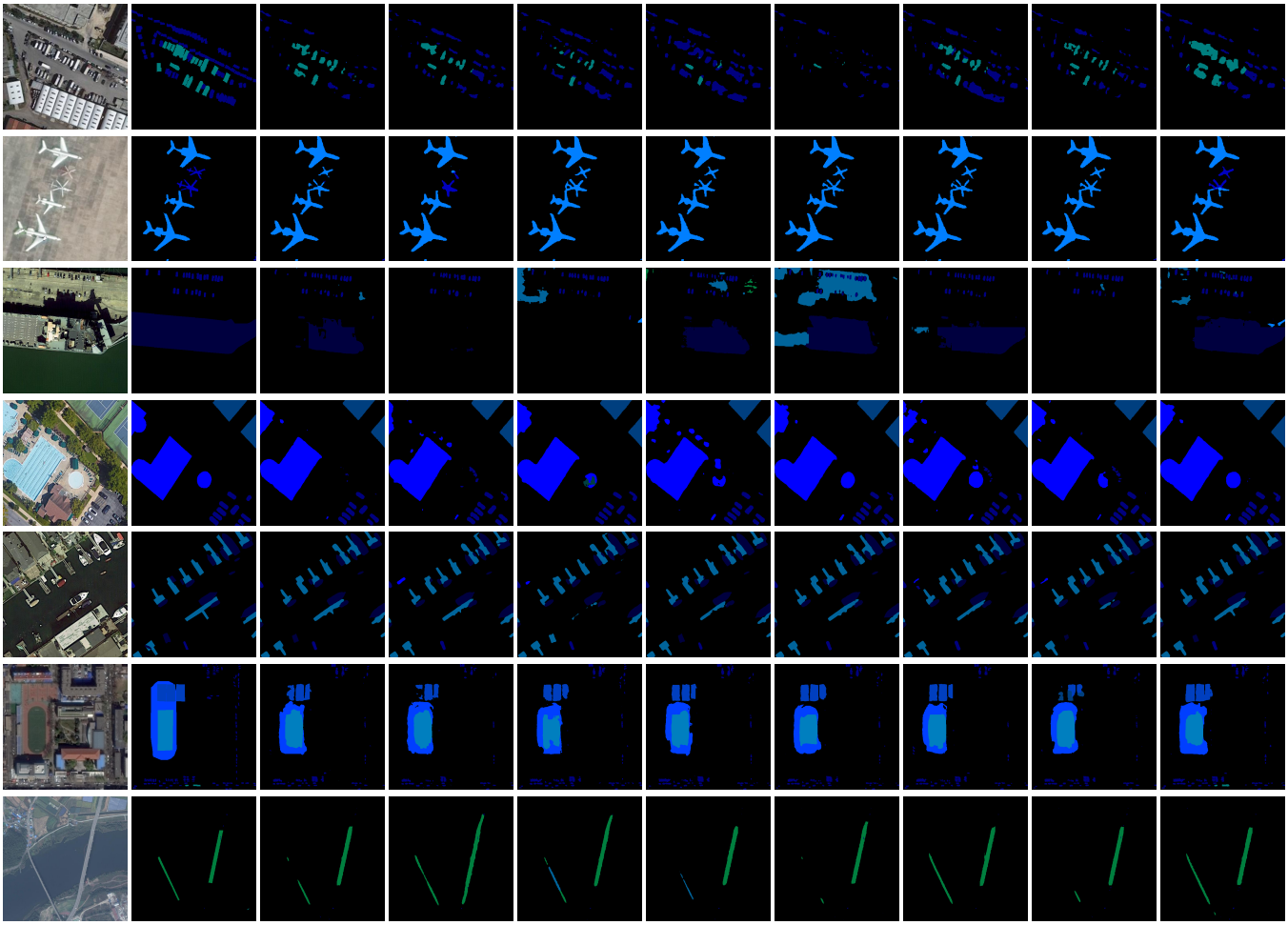

Fig. 5. Qualitative semantic segmentation results on iSAID validation set. (a) Image. (b) Ground truth label. (c) FPN. (d) FarSeg. (e) PFSegNet. (f) UNetFormer. (g) SCO. (h) MPCNet. (i) LOGCAN++. (j) DSA-Net.

classifiers on the same feature extraction network ConvNeXt [84], similar to [85]. The FLOPs, FPS, and memory consumption are measured under one 1024×1024 input image to the networks under the evaluation mode. The CenterSeg [85] classifier takes into account the dynamic classification prototypes for each patch of an image. However, the quadratic computation complexity in passing the dynamic prototype features to static prototype features increases the FLOPs and slows down the inference speed of the model, and the sample information for each class can be potentially deficient in a local patch without any effective mechanisms for augmenting the embedded features. Although the inference speed of MPC-Net is higher than DSA-Net, it is hard to train MPC-Net with more FLOPs, the requirement on memory banks of pixel features, and additional complex loss functions. DSA-Net achieves better accuracy and competitive inference speed with little additional parameters and FLOPs.

4) Qualitative Results: For qualitatively comparing different semantic segmentation methods, the visualized semantic segmentation results are presented in Figs. 5 and 6 with images on the validation sets of the iSAID dataset and the LoveDA dataset, respectively. The geospatial instances in complex scenarios are prone to ambiguities among different categories. It can be seen from Fig. 5 that not only the interference from the background, but also the misclassification of a geospatial object, appears when recognizing it as an incorrect foreground class. However, the proposed DSA-Net behaves more effectively in accurately distinguishing different classes in each scene, which can be observed from the relationships between classes such as planes and helicopters, small and large vehicles, or different types of large chunks of land-use parcels. The examples in the first and second row of Fig. 5 clearly demonstrates the cases of recognizing neighboring geospatial objects of easily ambiguous classes, in which DSA-Net better reflects the semantic information of small vehicles and helicopters and avoids misclassification. From the results in rows 3, 5, and 6, it can also be seen that the DSA-Net can segment the objects from the background in each scene more completely. For the results on the LoveDA dataset, DSA-Net shows better perceiving and distinguishing on different land-cover types, making it more adapted to the occurrence of relevant classes under each scene, which can be seen from the barren type in the third row and the road, agricultural, and building classes in the fourth row in

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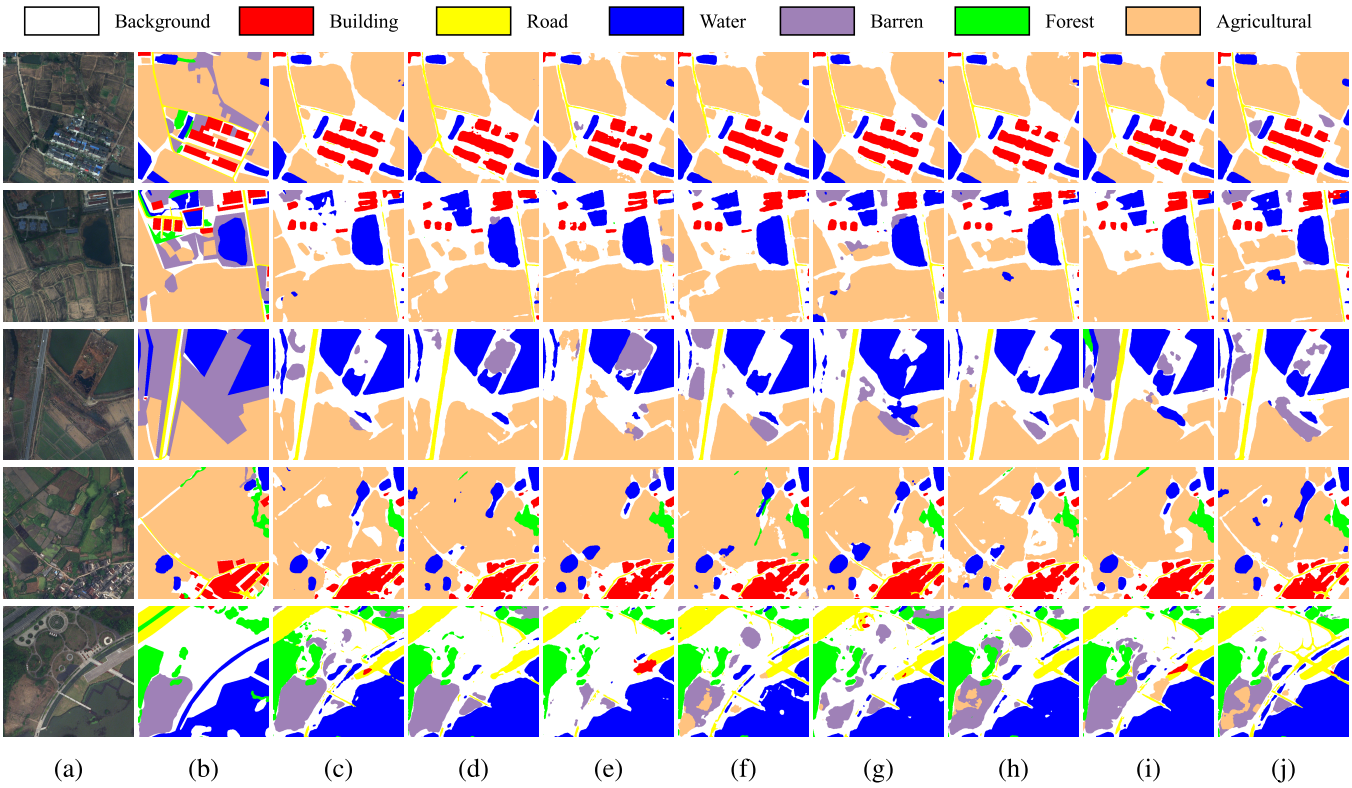

Fig. 6. Qualitative semantic segmentation results on LoveDA validation set. (a) Image. (b) Ground truth label. (c) FPN. (d) FarSeg. (e) PFSegNet. (f) UNetFormer. (g) SCO. (h) MPCNet. (i) LOGCAN++. (j) DSA-Net.

TABLE VII
EFFECTIVENESS OF THE PROPOSED METHOD EQUIPPED WITH DIFFERENT NETWORKS

Method	SH	ST	BD	TC	BC	GTF	BR	LV	SV	HC	SP	RA	SBF	PL	HA	mIoU (%)
FarSeg[12]	67.17	69.66	77.13	**89.88**	63.87	47.02	**37.65**	**63.26**	50.48	**35.02**	50.40	**65.87**	73.36	85.03	57.92	64.55
FarSeg + DSA-Net	**70.88**	**71.11**	**77.89**	89.59	**64.56**	**56.08**	34.99	63.14	**53.30**	32.78	**51.15**	64.61	**74.64**	**86.03**	**59.42**	**65.58**
UNetFormer [16]	66.25	69.15	79.05	88.78	59.34	**59.25**	37.55	60.77	47.44	13.60	**50.98**	67.45	**75.63**	82.95	53.00	63.14
UNetFormer + DSA-Net	**68.89**	**72.47**	**79.76**	**89.34**	**64.49**	58.73	**40.50**	**62.46**	**50.43**	**29.82**	49.74	**72.67**	73.23	**84.07**	**55.63**	**65.71**
FT-UNetFormer [16]	**74.00**	74.65	80.35	90.07	65.58	**59.64**	**42.08**	65.69	52.19	41.16	**55.85**	**71.48**	**75.23**	86.20	**58.60**	68.25
FT-UNetFormer + DSA-Net	73.28	**74.73**	**82.58**	**90.12**	**69.81**	59.51	41.72	**66.04**	**53.31**	**42.43**	54.96	68.27	74.14	**86.59**	58.56	**68.45**
PFSegNet [17]	69.77	73.33	82.03	**89.12**	**70.38**	**60.20**	**37.58**	61.89	**55.46**	27.07	**53.13**	65.00	**74.31**	85.80	59.83	66.50
PFSegNet + DSA-Net	**71.67**	**74.07**	**82.42**	88.87	64.32	60.08	36.29	**62.43**	54.09	**32.76**	52.56	**72.51**	70.92	**86.07**	**60.86**	**66.82**
LOGCAN++ [77]	68.79	**75.36**	79.94	88.41	64.16	**55.05**	**41.81**	64.20	**53.48**	**42.09**	53.68	62.58	**75.05**	**86.93**	**60.96**	66.98
LOGCAN++ + DSA-Net	**71.59**	75.12	**80.44**	**90.50**	**69.11**	50.18	40.19	**64.23**	52.35	39.80	**56.14**	**69.02**	72.74	86.89	59.55	**67.31**

Fig. 6 for examples. These results indicate that the proposed DSA-Net demonstrates better ability in recognizing and segmenting diverse classes on various remote sensing scenes. In Appendix B-C, the semantic segmentation results using the static prototype, the dynamic static prototype, and the final updated prototype for classification independently are presented, which indicate how the static and dynamic prototypes collaboratively participate in the classification decision process intuitively.

E. Ablation Study

1) Effectiveness of the Proposed Method as a Plugged-In Network Module: Table VII presents the semantic segmentation performance when equipping the proposed DSA-Net method on different models. To verify its effectiveness, the final classifier of these models is replaced by the proposed method with the proposed training approach. It can be seen that for different networks with different encoding-decoding computational flows, DSA-Net demonstrates beneficial improvement to these models. The overall mIoU gets improvement from 0.3% to 1.6% on these methods, indicating the effectiveness of DSA-Net on multiple semantic segmentation networks. Not only convolutional neural networks with certain special modules can benefit from DSA-Net, but the Transformer architectures, such as UNetFormer or Swin-B models, also get better performance on the OA and a majority of categories of the geospatial objects.

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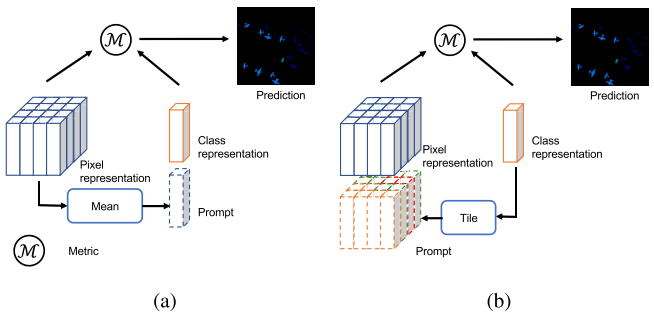

Fig. 7. Different strategies for collecting the scene knowledge. (a) Update the class representation and (b) update the pixel representation. The class representation or pixel representation is updated by first concatenating the prompt and the representation vectors, and using a learnable linear layer to project the concatenated vectors to the original dimension of the representation. The metric is the distance function to get the classification score of each pixel.

TABLE VIII
EFFECTIVENESS OF MODULE COMBINATION

$L_{\text{Final-CE}}$	$L_{\text{intra-class}}$	$L_{\text{inter-class}}$	Scene-Adaptive Prototypes	mIoU
✓				68.01
✓	✓			68.19
✓		✓		68.35
✓	✓	✓		68.56
✓			✓	69.26
✓	✓		✓	69.34
✓		✓	✓	69.67
✓	✓	✓	✓	**69.97**

2) Effectiveness of Module Combinations: We report the OA on the iSAID dataset with the DSA-Net applied to the ResNet-101 backbone to verify the effectiveness of the module combination in the proposed method in Table VIII. The basic static prototype layer obtained moderate performance on the semantic segmentation accuracy, verifying the basic feasibility of using distances to learnable classification prototypes for yielding the class-specific scores. Through adding the manifold mixup for augmenting the training data by filling or expanding the embedding space, the segmentation accuracy of the model improves step-by-step. The interclass embedding mixup seems more effective as a standalone approach compared with the intraclass embedding mixup for promoting the segmentation accuracy, probably because the learned Euclidean-distance-based embedding space is not mingled with complex interclass relationships. From this argument, we suggest designing with exploration on the shapes of localized embedding spaces if more fine-grained [22] or narrower [85] class-specific embedding spaces are pursued. When equipped with the scene-adaptive dynamic prototypes, the segmentation accuracies of different module combination cases are generally. The prototype and pixel representations in the learned embedding space of the DSA-Net are easy to interact with each other, thus making the combination of the prototype enhancement and the pixel augmentation easy to carry out. In Appendix B, we also conduct experiments on selecting pixels for extracting dynamic prototypes with different classification confidence thresholds, and evaluate the influence of intraclass and interclass manifold mixup.

3) Scene-Adaptive Dynamic Prototype Modules: To further explore the specific settings that influence the performance of the scene-adaptive dynamic prototype layer, we conducted a series of module setup toggling experiments in Table IX. The experiments were carried out using the same model as in Section IV-E2. Besides the accuracy of the semantic segmentation results, we also reported the training speed and training GPU memory occupation, as these factors are crucial to be controlled for constraining the operation complexities contained in a model. It can be observed from Table IX that using the hard class occurrence constraint operation [as (7)] is necessary to reduce unreliable classification prediction for updating the prototypes. The adoption of dynamic prototypes within each image is also important for effectively training the updated prototype layers, compared with using the dynamic prototypes across all images in a training batch. This indicates that the scene-adaptive information should be restricted in the localized view, while connecting different scenes globally may bring in undesired interference from irrelevant class information. The choice of guiding mask G for training is recommended to be the ground truth mask Y instead of the prediction mask $\tilde{M}$, which indicates that the scene-adaptive prototypes are potentially helpful for rectifying some incorrectly classified pixels in each scene.

4) Manifold Mixup Modules: Similar to the ablation study in Section IV-E3, Table X reports the accuracy and efficiency performances on the specific settings of the manifold mixup modules. The choices of cross-image or within-image interactions are compared, and we observed that the cross-image interaction is slightly better than the within-image interaction for interclass sample mixup. We suppose that the behaviors of distinguishing different classes are encouraged in the final classifier stage of a neural network. Thus, it is beneficial to take samples across all classes to drag or pull the classification prototypes into optimized locations. The parameters trained by the mixed embedded samples are recommended to be confined to the static prototype layer θ, which sees the embedding space itself for mixup directly. In comparison, the feature extractor ϕ and the scene-adaptive updated prototype layer θ' can be biased differently on the unlearned interclass regions in the embedding space.

V. DISCUSSION

To gain a deeper understanding of the dynamically updated class representations and further explore the mechanism of updating representations in semantic segmentation tasks, we carried out a series of experiments on the variants of the proposed model.

A. Dynamically Update Pixel Representation in Each Scene

A question of how to collect the scene knowledge is whether to update the pixel representation or update the class representation. As illustrated by Fig. 7, we can extract prompts to update class representation from the pixel representation in each scene, or we can use the class representation as

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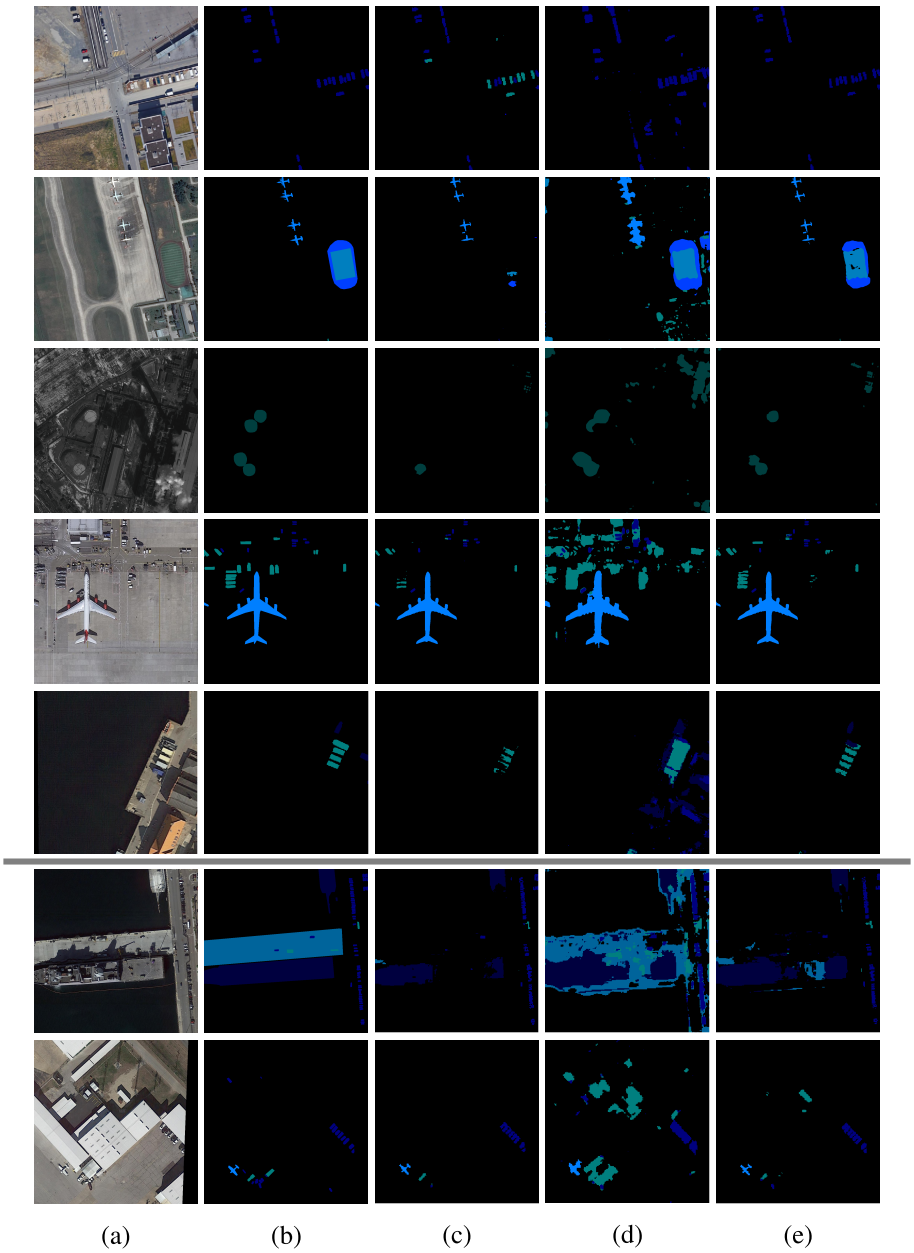

Fig. 8. Semantic segmentation results using the static, dynamic, and final updated prototypes for classification independently. (a) Image. (b) Ground truth label. (c) Static prototypes. (d) Dynamic prototypes. (e) Final updated prototypes. The last two rows are failure cases.

TABLE IX
SCENE-ADAPTIVE DYNAMIC PROTOTYPE MODULES

Module	Specification	(recommended)	Training speed (s/iteration)	Training memory (GB)	mIoU
Proposed model			0.4474	15.19	**69.26**
Scope of dynamic prototypes for training	batch	image	0.4143	15.14	68.45
Guiding mask G for training	$\tilde{M}$	Y	0.5988	15.46	68.65
Guiding mask G for inference	$\arg\max(\tilde{M})$	$\text{softmax}(\tilde{M})$	-	-	68.79
Hard class occurrence constraint	not use	use	0.4216	15.10	67.29

prompts to update the pixel representation. Both mechanisms make a combination between the static class representation and the dynamic pixel representation through a learnable linear projection, thus setting a scene-adaptive inference process into the semantic segmentation model. We emphasize that the pixel representation updating mechanism provides a wider adjustment space for the model inference process. We compared the accuracy and efficiency of the two mechanisms on the iSAID dataset in Table XI, and drew the following conclusions.

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TABLE X
MANIFOLD MIXUP MODULES

Module	Specification	(recommended)	Training speed (s/iteration)	Training memory (GB)	mIoU
Proposed model			0.3749	15.04	**69.35**
Image relationships in mixup	within-image	cross-image	0.3889	14.83	68.90
Trained modules	θ, ϕ	θ	0.3871	14.95	68.12
	θ'		0.3920	15.05	69.01
Classification loss re-weighting	not use	use	0.3745	14.97	49.65

1) The pixel representation update strategy does not have advantages in accuracy gain compared with the class prototype update strategy.
2) There is a noticeable computational efficiency decrease when using the pixel representation update strategy, compared with the class prototype update strategy.

From the above observations, although the pixel representation updating mechanism provides a wider adjustment range for the model inference process, designing an appropriate pixel representation updating solution is nontrivial. The handling of different pixels may contribute to more complicated dividing than their classification labels, which influences the advantages of the pixel representation update mechanism over the class representation update mechanism in practice. We describe the details of the pixel representation model in Appendix A.

TABLE XI
COMPARISON OF CLASS REPRESENTATION AND PIXEL REPRESENTATION UPDATE STRATEGIES

Method	Accuracy (mIoU)	Inference speed (FPS)	Training memory (GB)
Update class representation	69.26	161.16	16.29
Update pixel representation	68.87	123.98	16.73

TABLE XII
COMPARISON OF DIFFERENT CLASS EMBEDDING

Method	Accuracy (mIoU)
Static prototypes	68.87
Dynamic prototypes	68.35
Learnable class embedding	68.99

B. More Encoding for Pixel Prompts: Taking Pixel Classification Confidence in Pixel Prompt as an Example

Since the pixel representations are more abundant than the class representations, we can try to inject more knowledge into the prompts for dynamically updating the pixels. Here, we constructed a multiprompt mechanism for combining a learnable embedding for pixel classification confidence levels into the pixel representation. Unfortunately, experiments showed that updating the pixel representation with confidence values caused a significant mIoU accuracy drop (69.26 $\rightarrow$ 68.05). We summarized that simply superimposing more types of pixel prompts cannot lead to a better solution for semantic segmentation, and a more elaborate pixel prompt strategy might be required in practice.

C. Class-Specific Metrics

Adjustment to the class prototypes affects the class-wise measure of each pixel feature. However, a direct update to the relation between the pixel feature and the class prototypes has not been considered by either the class or pixel representation update models. To further explore the class-specific modeling of the classifier, we equipped the DSA-Net with learnable class-specific metrics based on the Mahalanobis distance, which plays the role of modeling the pixel-class relations. Surprisingly, this model achieved a higher segmentation accuracy on the iSAID dataset than the original DSA-Net. This finding indicates an effective way to explore the class-wise knowledge in addition to dynamically updating the class prototypes. From this observation, we argue that the class-wise information has a significant positive influence on the performance of semantic segmentation models. Focusing on the local geometry of the class prototypes may create better classification boundaries for each sample. We leave the detailed description and the experiment results of the class-specific metrics in Appendix C.

VI. CONCLUSION

In this article, we consider the problem of diversity of scenes and the heterogeneity of instances in the Earth observation data, and propose the DSA-Net to cope with these challenges in the semantic segmentation of remote sensing images. Specifically, the scene-adaptive prototype update mechanism is proposed, which combines the static class representation from the model and the dynamic class representation from the image features to dynamically fit various scenes. A prototype-based embedding space is constructed to facilitate the combination of static and dynamic class representations. A manifold mixup method is further proposed for training the model with intraclass and interclass diverse samples. Experiment results on four remote sensing semantic segmentation datasets verify the effectiveness of the proposed scene-adaptive classifier and the training approach. By observing the segmentation results, it is revealed that DSA-Net tends to activate more pixels with classes relevant to the current image, which can benefit the extraction of objects in scenes such as urban or industrial regions. There are still limitations that DSA-Net may be misled by some objects that have highly heterogeneous appearances, e.g., ships, docks, or large vehicles with intricate appearances. In our future work, we will further explore prior knowledge about the context of the geospatial objects that is able to guide

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TABLE XIII
DIFFERENT CONFIDENCE THRESHOLDS FOR MASKING PIXELS IN GENERATING DYNAMIC PROTOTYPES

Confidence threshold	SH	ST	BD	TC	BC	GTF	BR	LV	SV	HC	SP	RA	SBF	PL	HA	mIoU (%)
0.4	73.70	73.24	78.24	90.64	68.08	63.24	**42.16**	**65.94**	**54.50**	41.64	**54.26**	57.23	76.41	86.98	61.72	67.95
0.5	**74.19**	75.44	83.09	**90.89**	67.19	**64.51**	41.26	65.24	52.09	**45.50**	54.25	60.49	75.73	**87.01**	**63.54**	**68.73**
0.6	74.02	**75.83**	81.90	90.64	68.80	62.29	40.67	64.48	52.48	42.90	53.20	69.44	70.08	86.64	62.74	68.45
0.7	73.39	73.58	82.41	90.30	68.18	64.20	41.53	64.90	51.89	41.06	46.20	**70.20**	76.07	86.20	61.33	68.16
0.8	73.35	72.02	**83.12**	90.74	**71.39**	60.24	37.39	64.86	50.75	41.69	43.23	65.27	74.85	86.83	61.47	67.27
0.9	73.28	71.43	80.71	90.44	68.68	62.11	39.84	64.08	47.38	33.46	50.73	62.79	**76.42**	86.27	60.75	66.72

the classification model to adapt to diversified remote sensing scenes.

APPENDIX A
DETAILS OF PIXEL REPRESENTATION UPDATE MODEL

We describe the details of the class representation or pixel representation update models in Fig. 7 in this section.

A. Tiling Operation

The "tiling" operation in Fig. 7(b) is a process that embeds the class index of a pixel into a class-specific vector. Denote a class embedding as $E \in \mathbb{R}^{K\times C}$, where K is the number of classes and C is the latent dimension. A pixel classified as class k is then concatenated with a class-specific vector E_k, and yields a new pixel feature through a linear project ψ

$$x' = \psi\left([E_k, x]\right). \tag{11}$$

The class index of each pixel is from the coarse segmentation mask $\tilde{M}$ predicted by the model in our experiments.

B. Types of Class Embedding

There are multiple choices of the class embedding E. We selected static prototypes θ, dynamic prototypes p, and an independent learnable class embedding as the class-wise embeddings, and compared the performance on the iSAID dataset, as shown in Table XII. It can be seen from the experiment that changing the class embedding types is not a major factor that influences the segmentation results.

APPENDIX B
OTHER EXPERIMENTS

A. Selecting Pixels With Different Confidence Thresholds for Generating Dynamic Prototypes

The predicted probability of pixel classification is used as the confidence map, and is binarized with thresholds {0.4, 0.5, 0.6, 0.7, 0.8, 0.9} to select pixels for generating the dynamic prototypes from image features. If the threshold is greater than 0.5, high-confident pixels are chosen. If the threshold is less than 0.5, more ambiguously classified pixels are chosen for generating the dynamic prototypes. Table XIII reports the influence of selecting different confidence thresholds. The experiments support choosing 0.5 as the confidence threshold, while the accuracy drops more remarkably when the confidence threshold shifts farther away from 0.5.

B. Manifold Mixup Effectiveness

We report quantitative experiment results of the intraclass and interclass manifold mixup on each class and the overall accuracy on the iSAID dataset in Table XIV. The per-class performance of the two mixup types shows a positive effect on most of the classes.

C. Segmentation Results of Static Prototypes, Dynamic Prototypes, and Final Updated Prototypes

The semantic segmentation results using the static prototype, the dynamic prototype, and the final updated prototype for classification independently are presented in Fig. 8. It can be observed that there are noticeable differences among the segmentation results of the three types of classifiers. Generally, the segmentation results from the static prototypes and the final updated prototypes are close to the ground truth masks. However, the final updated prototypes tend to activate more pixels with classes relevant to the current image. Although the segmentation results yielded by the dynamic prototypes do not strictly coincide with the ground truth masks, they make the similar pixels on classes relevant to the current image closer to the final updated prototypes. Failure cases reveal that DSA-Net may be misled by some objects that have highly heterogeneous appearance, e.g., ships, docks, or large vehicles with intricate appearance.

APPENDIX C
CLASS-SPECIFIC METRICS

Updating the class prototypes creates class-specific distance spaces for the pixel feature. As shown in the following formula, when updating the prototype θ_k to $(\theta_k + \Delta\theta_k)$ for class k, the relation between pixel feature $x \in \mathbb{R}^C$ to the updated class prototype satisfies:

$$x - (\theta_k + \Delta\theta_k) = (x - \Delta\theta_k) - \theta_k. \tag{12}$$

The right-hand side of (12) can be explained as updating the pixel feature from x to $(x - \Delta\theta_k)$, which is a class-specific transformation instead of a class-agnostic transformation.

Motivated by this, we further assume that the relations between the pixel feature and the class prototypes can be modeled in a class-specific way. The mechanism of learning class-specific relations can be attributed to learning multiple metrics. We take the commonly used learnable Mahalanobis distance to build our classifier with class-specific metrics

$$f_k(x;\theta) = -\frac{1}{C}\|\theta_k - x\|^2_{L_k} = -\frac{1}{C}(\theta_k - x)^T L_k^T L_k (\theta_k - x) \tag{13}$$

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TABLE XIV
INFLUENCE OF INTRACLASS AND INTERCLASS MANIFOLD MIXUP

Method	SH	ST	BD	TC	BC	GTF	BR	LV	SV	HC	SP	RA	SBF	PL	HA	mIoU (%)
Baseline	73.83	75.32	81.11	90.62	66.38	**63.10**	42.10	66.57	**55.98**	42.26	**55.65**	**70.54**	75.28	87.14	63.08	69.26
Intra-class mixup	74.15	**76.39**	81.42	90.74	**69.27**	62.92	40.73	66.81	55.65	44.58	53.96	67.70	**75.29**	**87.28**	**63.37**	69.34
	(+0.32)	(+1.07)	(+0.31)	(+0.12)	(+2.89)	(-0.18)	(-1.37)	(+0.24)	(-0.33)	(+2.32)	(-1.69)	(-2.84)	(+0.01)	(+0.14)	(+0.29)	(+0.08)
Inter-class mixup	**74.23**	75.33	**82.83**	**90.75**	68.92	62.84	**43.21**	**66.95**	55.77	**48.23**	53.10	68.56	75.19	87.25	62.44	**69.67**
	(+0.40)	(+0.01)	(+1.72)	(+0.13)	(+2.54)	(-0.26)	(+1.11)	(+0.38)	(-0.21)	(+5.97)	(-2.55)	(-1.98)	(-0.09)	(+0.11)	(-0.64)	(+0.41)

TABLE XV
INFLUENCE OF CLASSIFIERS WITH CLASS-SPECIFIC METRICS

Method	SH	ST	BD	TC	BC	GTF	BR	LV	SV	HC	SP	RA	SBF	PL	HA	mIoU (%)
DSA-Net	72.82	76.29	83.51	**90.78**	65.99	**65.96**	41.67	**66.76**	55.94	**46.01**	**56.11**	71.91	76.63	87.22	62.77	69.97
+ class-specific metrics	**75.13**	**76.79**	**83.94**	90.71	**69.53**	64.71	**42.30**	66.13	**56.03**	41.02	55.88	**72.11**	**77.11**	**87.27**	**64.38**	**70.14**

where f_k is the classification score of class k for the pixel feature x. $L_k \in \mathbb{R}^{C\times C}$ is a learnable matrix, and $L_k^T L_k$ is the covariance matrix for computing the Mahalanobis distance specified for class k. By replacing the original classification function in (3) with (12), we get the variant model using a classifier equipped with the class-specific metrics. Table XV reports the model performance on the iSAID dataset using ResNet-101 as the backbone, when equipping the classifier with class-specific metrics.

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